MPG No. 4.1.9 m [Type I, monoclinic] [M tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & M_y & 0 \end{bmatrix}$$

$$M_y = \mathbb{M}_y^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ M_{xy} & 0 & M_{yz} \\ 0 & M_{yz} & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{M}_{xy}^{(1)}$$
$$M_{yz} = \mathbb{M}_{yz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & M_{yz} \\ 0 & -M_{yz} & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{T}_z^{(1)}$$
$$M_{yz} = \mathbb{T}_x^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & M_{xxy} & 0 \\ 0 & M_{yyy} & 0 \\ 0 & M_{zzy} & 0 \\ M_{yzx} & 0 & M_{zzy} \\ 0 & M_{yzx} & 0 \\ M_{xxy} & 0 & M_{yzx} \end{bmatrix}$$

$$M_{xxy} = \mathbb{M}_y^{(1)} - \mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\beta(1)}$$
$$M_{yyy} = 3\mathbb{M}_y^{(1)} + 2\mathbb{M}_y^{\alpha(1)}$$
$$M_{zzy} = \mathbb{M}_y^{(1)} - \mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\beta(1)}$$
$$M_{yzx} = \mathbb{M}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & M_{zzy} & 0 \\ M_{yzx} & 0 & -\frac{M_{zzy}}{2} \\ 0 & M_{zxy} & 0 \\ -\frac{M_{xxy}}{2} & 0 & -M_{yzx} - M_{zxy} \end{bmatrix}$$

$$M_{xxy} = 2\mathbb{M}_y^{(2)} - 2\mathbb{T}_{xz}^{(1)}$$
$$M_{zzy} = 2\mathbb{M}_y^{(2)} + 2\mathbb{T}_{xz}^{(1)}$$
$$M_{yzx} = 3\mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$
$$\frac{M_{zzy}}{2} = \mathbb{M}_y^{(2)} + \mathbb{T}_{xz}^{(1)}$$
$$M_{zxy} = -3\mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$
$$\frac{M_{xxy}}{2} = \mathbb{M}_y^{(2)} - \mathbb{T}_{xz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & M_{yzz} \\ 0 & M_{zxy} & 0 \\ M_{xyx} & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)} + \mathbb{T}_v^{(2)}$$
$$M_{yzz} = -\mathbb{M}_y^{(3)} + \mathbb{T}_{xz}^{(2)}$$
$$M_{zxy} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)} - \mathbb{T}_v^{(2)}$$
$$M_{xyx} = \mathbb{M}_y^{(3)} + \mathbb{T}_{xz}^{(2)}$$
$$M_{xyz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & M_{xxxy} \\ 0 & 0 & 0 & M_{yyyz} & 0 & M_{yyxy} \\ 0 & 0 & 0 & M_{zzyz} & 0 & M_{zzxy} \\ M_{xxyz} & M_{yyyz} & M_{zzyz} & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & 0 & M_{zxxy} \\ M_{xxxy} & M_{yyxy} & M_{zzxy} & 0 & M_{zxxy} & 0 \end{bmatrix}$$

$$M_{xxyz} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)}$$
$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)}$$
$$M_{yyyz} = \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)}$$
$$M_{yyxy} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)}$$
$$M_{zzyz} = -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)}$$
$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)}$$
$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)}$$
$$M_{zxxy} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & M_{xxxy} \\ 0 & 0 & 0 & M_{yyyz} & 0 & M_{yyxy} \\ 0 & 0 & 0 & M_{zzyz} & 0 & M_{zzxy} \\ -M_{xxyz} & -M_{yyyz} & -M_{zzyz} & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & M_{zxxy} \\ -M_{xxxy} & -M_{yyxy} & -M_{zzxy} & 0 & -M_{zxxy} & 0 \end{bmatrix}$$

$$M_{xxyz} = \mathbb{M}_{yz}^{(3)} - 4\mathbb{T}_x^{\beta(1)}$$
$$M_{xxxy} = \mathbb{M}_{xy}^{(3)} + 2\mathbb{T}_z^{(1)} - 2\mathbb{T}_z^{\alpha(1)} + 2\mathbb{T}_z^{\beta(1)}$$
$$M_{yyyz} = \mathbb{M}_{yz}^{(3)} + 2\mathbb{T}_x^{(1)} - 2\mathbb{T}_x^{\alpha(1)} + 2\mathbb{T}_x^{\beta(1)}$$
$$M_{yyxy} = \mathbb{M}_{xy}^{(3)} - 2\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)} + 2\mathbb{T}_z^{\beta(1)}$$
$$M_{zzyz} = \mathbb{M}_{yz}^{(3)} - 2\mathbb{T}_x^{(1)} + 2\mathbb{T}_x^{\alpha(1)} + 2\mathbb{T}_x^{\beta(1)}$$
$$M_{zzxy} = \mathbb{M}_{xy}^{(3)} - 4\mathbb{T}_z^{\beta(1)}$$
$$M_{yzzx} = -\mathbb{T}_z^{(1)} - 4\mathbb{T}_z^{\alpha(1)}$$
$$M_{zxxy} = -\mathbb{T}_x^{(1)} - 4\mathbb{T}_x^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ M_{yzzx} & 0 & M_{zxxy} \\ 0 & M_{zxxy} & 0 \end{bmatrix}$$

$$M_{yzzx} = 2\mathbb{M}^{(4)}_{xy}$$
$$M_{zxxy} = 2\mathbb{M}^{(4)}_{yz}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ -M_{yzzx} & 0 & M_{zxxy} \\ 0 & -M_{zxxy} & 0 \end{bmatrix}$$

$$M_{yzzx} = -\mathbb{T}^{(2)}_{z}$$
$$M_{zxxy} = -\mathbb{T}^{(2)}_{x}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xxyz} & 0 & M_{xxxy} \\ M_{yyyz} & 0 & M_{yyxy} \\ M_{zzyz} & 0 & M_{zzxy} \\ 0 & M_{yzzx} & 0 \\ M_{zxyz} & 0 & M_{zxxy} \\ 0 & M_{xyzx} & 0 \end{bmatrix}$$

$$M_{xxyz} = \mathbb{T}^{(3)}_{x} + 2\mathbb{T}^{(4)}_{x} + 2\mathbb{T}^{\alpha(2)}_{x}$$
$$M_{xxxy} = -2\mathbb{M}^{(5)}_{xy} + \mathbb{T}^{(3)}_{z} - \mathbb{T}^{\alpha(2)}_{z} + \mathbb{T}^{\beta(2)}_{z}$$
$$M_{yyyz} = -2\mathbb{M}^{(5)}_{yz} + \mathbb{T}^{(3)}_{x} - \mathbb{T}^{\alpha(2)}_{x} + \mathbb{T}^{\beta(2)}_{x}$$
$$M_{yyxy} = 2\mathbb{M}^{(5)}_{xy} + \mathbb{T}^{(3)}_{z} - \mathbb{T}^{\alpha(2)}_{z} - \mathbb{T}^{\beta(2)}_{z}$$
$$M_{zzyz} = 2\mathbb{M}^{(5)}_{yz} + \mathbb{T}^{(3)}_{x} - \mathbb{T}^{\alpha(2)}_{x} - \mathbb{T}^{\beta(2)}_{x}$$
$$M_{zzxy} = \mathbb{T}^{(3)}_{z} + 2\mathbb{T}^{(4)}_{z} + 2\mathbb{T}^{\alpha(2)}_{z}$$
$$M_{yzzx} = -\mathbb{M}^{(5)}_{xy} + \mathbb{T}^{(4)}_{z} - \mathbb{T}^{\alpha(2)}_{z} - \mathbb{T}^{\beta(2)}_{z}$$
$$M_{zxyz} = \mathbb{M}^{(5)}_{xy} + \mathbb{T}^{(4)}_{z} - \mathbb{T}^{\alpha(2)}_{z} + \mathbb{T}^{\beta(2)}_{z}$$
$$M_{zxxy} = -\mathbb{M}^{(5)}_{yz} + \mathbb{T}^{(4)}_{x} - \mathbb{T}^{\alpha(2)}_{x} - \mathbb{T}^{\beta(2)}_{x}$$
$$M_{xyzx} = \mathbb{M}^{(5)}_{yz} + \mathbb{T}^{(4)}_{x} - \mathbb{T}^{\alpha(2)}_{x} + \mathbb{T}^{\beta(2)}_{x}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} M_{yzxx} & M_{yzyy} & M_{yzzz} & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{zxyz} & 0 & M_{zxxy} \\ M_{xyxx} & M_{xyyy} & M_{xyzz} & 0 & M_{xyzx} & 0 \end{bmatrix}$$

$$M_{yzxx} = \mathbb{T}^{(5)}_{x} + 2\mathbb{T}^{(6)}_{x} + 2\mathbb{T}^{\alpha(3)}_{x}$$
$$M_{yzyy} = -2\mathbb{M}^{(6)}_{yz} + \mathbb{T}^{(5)}_{x} - \mathbb{T}^{\alpha(3)}_{x} + \mathbb{T}^{\beta(3)}_{x}$$
$$M_{yzzz} = 2\mathbb{M}^{(6)}_{yz} + \mathbb{T}^{(5)}_{x} - \mathbb{T}^{\alpha(3)}_{x} - \mathbb{T}^{\beta(3)}_{x}$$
$$M_{yzzx} = \mathbb{M}^{(6)}_{xy} + \mathbb{T}^{(6)}_{z} - \mathbb{T}^{\alpha(3)}_{z} + \mathbb{T}^{\beta(3)}_{z}$$
$$M_{zxyz} = -\mathbb{M}^{(6)}_{xy} + \mathbb{T}^{(6)}_{z} - \mathbb{T}^{\alpha(3)}_{z} - \mathbb{T}^{\beta(3)}_{z}$$
$$M_{zxxy} = \mathbb{M}^{(6)}_{yz} + \mathbb{T}^{(6)}_{x} - \mathbb{T}^{\alpha(3)}_{x} + \mathbb{T}^{\beta(3)}_{x}$$
$$M_{xyxx} = -2\mathbb{M}^{(6)}_{xy} + \mathbb{T}^{(5)}_{z} - \mathbb{T}^{\alpha(3)}_{z} + \mathbb{T}^{\beta(3)}_{z}$$
$$M_{xyyy} = 2\mathbb{M}^{(6)}_{xy} + \mathbb{T}^{(5)}_{z} - \mathbb{T}^{\alpha(3)}_{z} - \mathbb{T}^{\beta(3)}_{z}$$
$$M_{xyzz} = \mathbb{T}^{(5)}_{z} + 2\mathbb{T}^{(6)}_{z} + 2\mathbb{T}^{\alpha(3)}_{z}$$
$$M_{xyzx} = -\mathbb{M}^{(6)}_{yz} + \mathbb{T}^{(6)}_{x} - \mathbb{T}^{\alpha(3)}_{x} - \mathbb{T}^{\beta(3)}_{x}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & M_{xxxy} & M_{xxzy} & 0 & M_{xxyx} \\ 0 & 0 & 0 & M_{yyyz} & 0 & M_{yyxy} & M_{yyzy} & 0 & M_{yyyx} \\ 0 & 0 & 0 & M_{zzyz} & 0 & M_{zzxy} & M_{zzzy} & 0 & M_{zzyx} \\ M_{yzxx} & M_{yzyy} & M_{yzzz} & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{zxyz} & 0 & M_{zxxy} & M_{zxzy} & 0 & M_{zxyx} \\ M_{xyxx} & M_{xyyy} & M_{xyzz} & 0 & M_{xyzx} & 0 & 0 & M_{xyxz} & 0 \end{bmatrix}$$

$$M_{xxyz} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + \mathbb{M}_{yz}^{(3)} + \mathbb{T}_{x}^{(3)} + 2\mathbb{T}_{x}^{(4)} + 2\mathbb{T}_{x}^{\alpha(2)} - 4\mathbb{T}_{x}^{\beta(1)}$$

$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} + 2\mathbb{T}_{z}^{(1)} + \mathbb{T}_{z}^{(3)} - 2\mathbb{T}_{z}^{\alpha(1)} - \mathbb{T}_{z}^{\alpha(2)} + 2\mathbb{T}_{z}^{\beta(1)} + \mathbb{T}_{z}^{\beta(2)}$$

$$M_{xxzy} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + \mathbb{M}_{yz}^{(3)} - \mathbb{T}_{x}^{(3)} - 2\mathbb{T}_{x}^{(4)} - 2\mathbb{T}_{x}^{\alpha(2)} - 4\mathbb{T}_{x}^{\beta(1)}$$

$$M_{xxyx} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} + 2\mathbb{T}_{z}^{(1)} - \mathbb{T}_{z}^{(3)} - 2\mathbb{T}_{z}^{\alpha(1)} + \mathbb{T}_{z}^{\alpha(2)} + 2\mathbb{T}_{z}^{\beta(1)} - \mathbb{T}_{z}^{\beta(2)}$$

$$M_{yyyz} = \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} - 2\mathbb{M}_{yz}^{(5)} + 2\mathbb{T}_{x}^{(1)} + \mathbb{T}_{x}^{(3)} - 2\mathbb{T}_{x}^{\alpha(1)} - \mathbb{T}_{x}^{\alpha(2)} + 2\mathbb{T}_{x}^{\beta(1)} + \mathbb{T}_{x}^{\beta(2)}$$

$$M_{yyxy} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} - 2\mathbb{T}_{z}^{(1)} + \mathbb{T}_{z}^{(3)} + 2\mathbb{T}_{z}^{\alpha(1)} - \mathbb{T}_{z}^{\alpha(2)} + 2\mathbb{T}_{z}^{\beta(1)} - \mathbb{T}_{z}^{\beta(2)}$$

$$M_{yyzy} = \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} + 2\mathbb{M}_{yz}^{(5)} + 2\mathbb{T}_{x}^{(1)} - \mathbb{T}_{x}^{(3)} - 2\mathbb{T}_{x}^{\alpha(1)} + \mathbb{T}_{x}^{\alpha(2)} + 2\mathbb{T}_{x}^{\beta(1)} - \mathbb{T}_{x}^{\beta(2)}$$

$$M_{yyyx} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - 2\mathbb{T}_{z}^{(1)} - \mathbb{T}_{z}^{(3)} + 2\mathbb{T}_{z}^{\alpha(1)} + \mathbb{T}_{z}^{\alpha(2)} + 2\mathbb{T}_{z}^{\beta(1)} + \mathbb{T}_{z}^{\beta(2)}$$

$$M_{zzyz} = -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} + 2\mathbb{M}_{yz}^{(5)} - 2\mathbb{T}_{x}^{(1)} + \mathbb{T}_{x}^{(3)} + 2\mathbb{T}_{x}^{\alpha(1)} - \mathbb{T}_{x}^{\alpha(2)} + 2\mathbb{T}_{x}^{\beta(1)} - \mathbb{T}_{x}^{\beta(2)}$$

$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} + \mathbb{T}_{z}^{(3)} + 2\mathbb{T}_{z}^{(4)} + 2\mathbb{T}_{z}^{\alpha(2)} - 4\mathbb{T}_{z}^{\beta(1)}$$

$$M_{zzzy} = -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} - 2\mathbb{M}_{yz}^{(5)} - 2\mathbb{T}_{x}^{(1)} - \mathbb{T}_{x}^{(3)} + 2\mathbb{T}_{x}^{\alpha(1)} + \mathbb{T}_{x}^{\alpha(2)} + 2\mathbb{T}_{x}^{\beta(1)} + \mathbb{T}_{x}^{\beta(2)}$$

$$M_{zzyx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} - \mathbb{T}_{z}^{(3)} - 2\mathbb{T}_{z}^{(4)} - 2\mathbb{T}_{z}^{\alpha(2)} - 4\mathbb{T}_{z}^{\beta(1)}$$

$$M_{yzxx} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} - \mathbb{M}_{yz}^{(3)} + 4\mathbb{T}_{x}^{\beta(1)}$$

$$M_{yzyy} = \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(3)} - 2\mathbb{T}_{x}^{(1)} + 2\mathbb{T}_{x}^{\alpha(1)} - 2\mathbb{T}_{x}^{\beta(1)}$$

$$M_{yzzz} = -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(3)} + 2\mathbb{T}_{x}^{(1)} - 2\mathbb{T}_{x}^{\alpha(1)} - 2\mathbb{T}_{x}^{\beta(1)}$$

$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} - \mathbb{T}_{z}^{(1)} + \mathbb{T}_{z}^{(4)} - 4\mathbb{T}_{z}^{\alpha(1)} - \mathbb{T}_{z}^{\alpha(2)} - \mathbb{T}_{z}^{\beta(2)}$$

$$M_{yzxz} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} - \mathbb{T}_{z}^{(1)} - \mathbb{T}_{z}^{(4)} - 4\mathbb{T}_{z}^{\alpha(1)} + \mathbb{T}_{z}^{\alpha(2)} + \mathbb{T}_{z}^{\beta(2)}$$

$$M_{zxyz} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} + \mathbb{T}_{z}^{(1)} + \mathbb{T}_{z}^{(4)} + 4\mathbb{T}_{z}^{\alpha(1)} - \mathbb{T}_{z}^{\alpha(2)} + \mathbb{T}_{z}^{\beta(2)}$$

$$M_{zxxy} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(5)} - \mathbb{T}_{x}^{(1)} + \mathbb{T}_{x}^{(4)} - 4\mathbb{T}_{x}^{\alpha(1)} - \mathbb{T}_{x}^{\alpha(2)} - \mathbb{T}_{x}^{\beta(2)}$$

$$M_{zxzy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} + \mathbb{T}_{z}^{(1)} - \mathbb{T}_{z}^{(4)} + 4\mathbb{T}_{z}^{\alpha(1)} + \mathbb{T}_{z}^{\alpha(2)} - \mathbb{T}_{z}^{\beta(2)}$$

$$M_{zxyx} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(5)} - \mathbb{T}_{x}^{(1)} - \mathbb{T}_{x}^{(4)} - 4\mathbb{T}_{x}^{\alpha(1)} + \mathbb{T}_{x}^{\alpha(2)} + \mathbb{T}_{x}^{\beta(2)}$$

$$M_{xyxx} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} - 2\mathbb{T}_{z}^{(1)} + 2\mathbb{T}_{z}^{\alpha(1)} - 2\mathbb{T}_{z}^{\beta(1)}$$

$$M_{xyyy} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} + 2\mathbb{T}_{z}^{(1)} - 2\mathbb{T}_{z}^{\alpha(1)} - 2\mathbb{T}_{z}^{\beta(1)}$$

$$M_{xyzz} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} - \mathbb{M}_{xy}^{(3)} + 4\mathbb{T}_{z}^{\beta(1)}$$

$$M_{xyzx} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(5)} + \mathbb{T}_{x}^{(1)} + \mathbb{T}_{x}^{(4)} + 4\mathbb{T}_{x}^{\alpha(1)} - \mathbb{T}_{x}^{\alpha(2)} + \mathbb{T}_{x}^{\beta(2)}$$

$$M_{xyxz} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(5)} + \mathbb{T}_{x}^{(1)} - \mathbb{T}_{x}^{(4)} + 4\mathbb{T}_{x}^{\alpha(1)} + \mathbb{T}_{x}^{\alpha(2)} - \mathbb{T}_{x}^{\beta(2)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} M_{yzxx} & M_{yzyy} & M_{yzzz} & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{zxyz} & 0 & M_{zxxy} & M_{zxzy} & 0 & M_{zxyx} \\ M_{xyxx} & M_{xyyy} & M_{xyzz} & 0 & M_{xyzx} & 0 & 0 & M_{xyxz} & 0 \end{bmatrix}$$

$$
\begin{aligned}
M_{yzxx} &= \mathbb{T}_x^{(5)} + 2\mathbb{T}_x^{(6)} + 2\mathbb{T}_x^{\alpha(3)} \\
M_{yzyy} &= -2\mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(5)} - \mathbb{T}_x^{\alpha(3)} + \mathbb{T}_x^{\beta(3)} \\
M_{yzzz} &= 2\mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(5)} - \mathbb{T}_x^{\alpha(3)} - \mathbb{T}_x^{\beta(3)} \\
M_{yzzx} &= 2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} - \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)} \\
M_{yzxz} &= -2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)} \\
M_{zxyz} &= 2\mathbb{M}_{xy}^{(4)} - \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)} \\
M_{zxxy} &= 2\mathbb{M}_{yz}^{(4)} + \mathbb{M}_{yz}^{(6)} - \mathbb{T}_x^{(2)} + \mathbb{T}_x^{(6)} - \mathbb{T}_x^{\alpha(3)} + \mathbb{T}_x^{\beta(3)} \\
M_{zxzy} &= -2\mathbb{M}_{xy}^{(4)} - \mathbb{M}_{xy}^{(6)} - \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)} \\
M_{zxyx} &= -2\mathbb{M}_{yz}^{(4)} + \mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(2)} + \mathbb{T}_x^{(6)} - \mathbb{T}_x^{\alpha(3)} + \mathbb{T}_x^{\beta(3)} \\
M_{xyxx} &= -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)} \\
M_{xyyy} &= 2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)} \\
M_{xyzz} &= \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)} \\
M_{xyzx} &= 2\mathbb{M}_{yz}^{(4)} - \mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(2)} + \mathbb{T}_x^{(6)} - \mathbb{T}_x^{\alpha(3)} - \mathbb{T}_x^{\beta(3)} \\
M_{xyxz} &= -2\mathbb{M}_{yz}^{(4)} - \mathbb{M}_{yz}^{(6)} - \mathbb{T}_x^{(2)} + \mathbb{T}_x^{(6)} - \mathbb{T}_x^{\alpha(3)} - \mathbb{T}_x^{\beta(3)}
\end{aligned}
$$

* Rank 4 tensor (t).

$$
\begin{bmatrix}
0 & M_{xxxy} & 0 \\
M_{yyyx} & 0 & M_{yyyz} \\
0 & M_{zzzy} & 0 \\
0 & M_{yyzy} & 0 \\
0 & M_{zzxy} & 0 \\
M_{xxyx} & 0 & M_{xxyz} \\
M_{yzzx} & 0 & M_{yzzz} \\
0 & M_{zxxy} & 0 \\
0 & M_{xyyy} & 0 \\
-\frac{M_{yyyz}}{2} + \frac{M_{yyzy}}{2} + M_{zxxy} & 0 & \frac{M_{xxxy}}{2} - \frac{M_{xxyx}}{2} + M_{xyyy} - M_{yyyx} + M_{zzxy}
\end{bmatrix}
$$

$$
\begin{aligned}
M_{xxxy} &= \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} \\
M_{yyyx} &= -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} \\
M_{yyyz} &= \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} - 2\mathbb{M}_{yz}^{(5)} \\
M_{zzzy} &= -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} - 2\mathbb{M}_{yz}^{(5)} \\
M_{yyzy} &= \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} + 2\mathbb{M}_{yz}^{(5)} \\
M_{zzxy} &= 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} \\
M_{xxyx} &= \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} \\
M_{xxyz} &= 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + \mathbb{M}_{yz}^{(3)} \\
M_{yzzx} &= 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} \\
M_{yzzz} &= -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(3)} \\
M_{zxxy} &= 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(5)} \\
M_{xyyy} &= -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)}
\end{aligned}
$$